QUESTION-5

Single linked list using insertion in beginning, end, at any position

main.c   Run

```c
#include <stdio.h>
#include <stdlib.h>
struct node {
    int data;
    struct node *next;
};
struct node *head = NULL;
void insertBegin(int value) {
    struct node *newNode = (struct node
        *)malloc(sizeof(struct node));
    newNode->data = value;
    newNode->next = head;
    head = newNode;
}
void insertEnd(int value) {
    struct node *newNode = (struct node
        *)malloc(sizeof(struct node));
    struct node *temp = head;
    newNode->data = value;
    newNode->next = NULL;
    if (head == NULL) {
        head = newNode;
        return;
    }
```

Output   Clear

```
Linked List: 5 -> 10 -> 15 -> 20 -> 30 -> NULL

=== Code Execution Successful ===
```

main.c   Run

```c
    while (temp->next != NULL)
        temp = temp->next;
    temp->next = newNode;
}
void insertPosition(int value, int pos) {
    struct node *newNode, *temp;
    int i;
    newNode = (struct node *)malloc(sizeof
        (struct node));
    newNode->data = value;
    if (pos == 1) {
        newNode->next = head;
        head = newNode;
        return;
    }

    temp = head;
    for (i = 1; i < pos - 1 && temp != NULL; i
        ++)
        temp = temp->next;

    if (temp == NULL) {
        printf("Invalid Position\n");
        return;
```

Output   Clear

```
Linked List: 5 -> 10 -> 15 -> 20 -> 30 -> NULL

=== Code Execution Successful ===
```

main.c   Run

```c
        return;
    }

    printf("Linked List: ");
    while (temp != NULL) {
        printf("%d -> ", temp->data);
        temp = temp->next;
    }
    printf("NULL\n");
}

int main() {
    insertBegin(10);
    insertBegin(5);

    insertEnd(20);
    insertEnd(30);

    insertPosition(15, 3);

    display();

    return 0;
}
```

Output   Clear

```
Linked List: 5 -> 10 -> 15 -> 20 -> 30 -> NULL

=== Code Execution Successful ===
```